

Research Structure to Evaluate the Child Marriage Ban in Brazil

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April 28, 2026

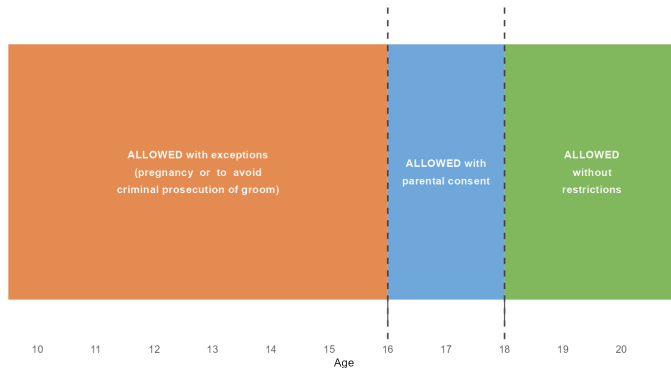
Overview

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- 5 Mechanisms: A 2×2 Typology
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- 7 Empirical Strategy
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Motivation

Before Lei 13.811/2019: a permissive regime below age 16

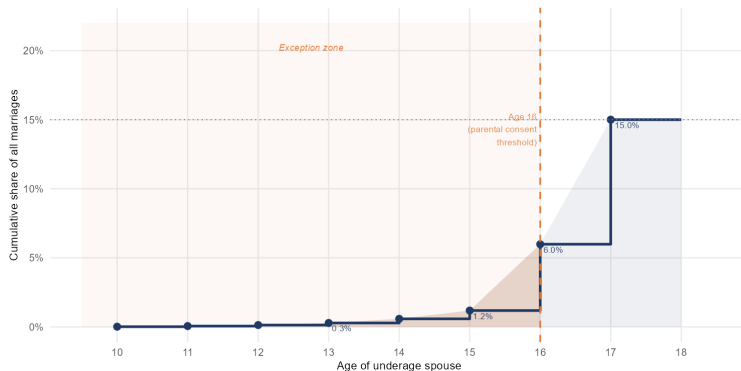
Art. 1.520 of the Civil Code allowed a judge to grant marriage *below age 16* in two circumstances: (i) pregnancy, or (ii) to avoid criminal prosecution of the adult.



Motivation

Baseline distribution of formal marriages before 2019

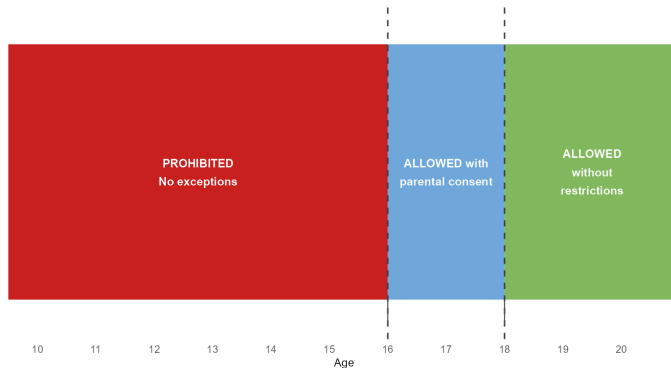
Under this regime, xx% of formal child marriages were concentrated at ages 16–17 between yyyy to 2019, but the exception clause generates a **small tail below age 16**.



Motivation

Lei 13.811/2019: an unconditional prohibition below age 16

Enacted 12 March 2019, the law eliminated *all exceptions*: formal marriage below age 16 is now **absolutely prohibited** under any circumstance.



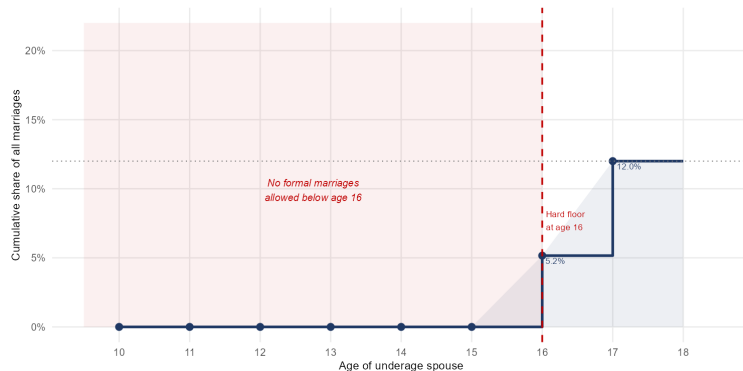
Hypothesis 1: Formal marriages below 16 should drop to zero

Under full enforcement, the Registro Civil should show **no** formal marriages below age 16 after 2019.

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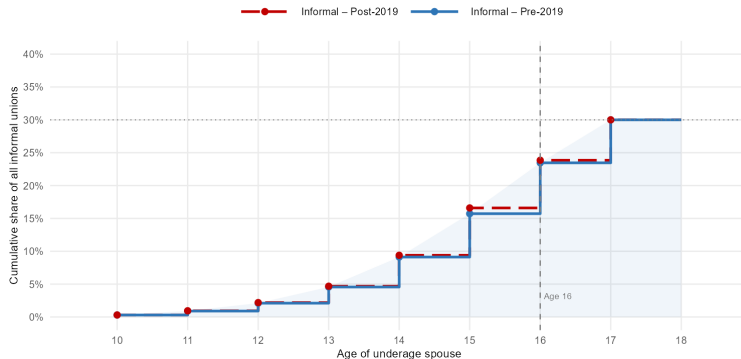


Hypothesis 2: Informal unions are outside the law's reach

The civil-registry ban covers *only* formal marriages. Informal unions — **89.5% of child unions** (Census 2010) — are legally unreachable. If families substitute from formal to informal channels, the informal distribution should be **unchanged or larger** after 2019.

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Did the 2019 ban actually reduce child marriage in Brazil, if not, which marriages persisted?

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Research Questions

Primary question

Did Lei 13.811/2019 reduce child marriage in Brazil, and through which margins: formal or informal marriage, for which ages and regions?

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Secondary questions

- 1 What **mechanisms** sustain child marriage in Brazil: economic pressure, normative pressure, or both?
- 2 Can we quantify a **social norm share** of child marriages that legal and economic interventions alone cannot reach?

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Secondary questions

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Contribution

First causal evaluation of Lei 13.811/2019 combining a marriage-market model, heterogeneous-mechanism estimation, and formal and informal margin comparison.

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Child Marriage in Brazil

Who

- Girls, ages 12–17 (mode 14–15)
- Black and brown (65–75% across studies)
- Lower household education and income
- Often pregnant at time of union formation

Where

- Rural North and Northeast
- Center-West emerging (Pinto 2024)
- Peripheries of major cities (Margarido 2021)

Why

- Control of female sexuality / family honor (Taylor & Fonseca 2015)
- Poverty / “exit option” (Veiga & Zanello 2020)
- Pregnancy “repair” union (Taylor & Fonseca 2015)
- Intergenerational transmission (Asadullah 2018, globally)

What the PBF Evidence Already Tells Us

VASCONCELOS & GRIEBELER (2017, 2023) use Propensity Score Matching on PNADC data:

- Bolsa Família **reduces child marriage** among poor and extremely poor girls.
- **Protective effect is concentrated at ages 16–18.**
- **Null effect at 14–15:** income transfers do not move the youngest unions.

Interpretation

The age-heterogeneity in PBF response is already evidence that a large share of the youngest marriages are driven by mechanisms other than economic pressure.

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Study	Context / Method	Key finding
Field & Ambrus (2008)	Bangladesh; menar- che IV	One year marriage delay increases 0.22 year school- ing
Dahl (2010)	USA	Early marriage increases 31 pp poverty for women
Bharadwaj (2015)	Mississippi	Law reform decreased 75% marriages, and increased 3% enrollment
Duflo et al. (2015)	Kenya	Education subsidy beats HIV education for delay
Wahhaj (2015)	Bangladesh	Reputational equilibrium sustains early marriage
Ashraf et al. (2016)	Indonesia; Zambia	School supply binds in bride-price groups only
Collin & Talbot (2016)	60+ countries	Most minimum-age laws poorly enforced
Chari et al. (2017)	India; menarche IV	One year of marriage delay increases 3.1% child school enrollment
Corno et al. (2019)	SSA and India	Bride price increases and dowry decreases under droughts
Blank et al. (2020)	USA	Administrative data overstates law effects
Chow & Vivalt (2022)	Ethiopia	Community interventions reduce marriage 4–7 pp
Wilson (2022)	17 countries	Age bans raise marriage age, first-birth age, employ- ment

Study	Data or Method	Key finding
Taylor & Fonseca (2015)	Pará/Maranhão	Honor, sexuality control, pregnancy trigger unions
Vasconcelos & Griebeler (2017)	PNADC and PSM	PBF protects girls in extreme poverty
Veiga & Zanello (2020)	Goiás	Constrained agency
Cargnin et al. (2021)	Rio Branco and SINAN	41.2% of 10–14 yr sexual-violence victims in union
Cardoso et al. (2022)	PNS	3.9% prevalence; intersectional determinants
Vasconcelos & Griebeler (2023)	PNADC and PSM	PBF effect concentrated at 16–18; null at 14–15
Margarido (2024)	Census and Registro Civil	655k unions; 45.4 formal marriages per day in 2021
Pinto et al. (2024)	SINASC	North and Center-West clusters child marriage
Aguar et al. (2025)	SINAN	3,681 marital rapes of girls under 14 years old

Contrasting the Two Literatures

Dimension	Global literature	Brazilian literature
Causal identification	Strong	Mostly descriptive
Economic mechanism	Bride price, dowry, income shocks	Poverty as exit option; PBF
Norm mechanism	Formal equilibrium models	Qualitative, ethnographic
Race and class	Rarely foregrounded	Intersectionality central
Agency	Girls as passive	Constrained-agency theory

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Mechanisms: the 2×2 Typology

	Low normative pressure	High normative pressure
High economic pressure		
Low economic pressure		

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	Low normative pressure	High normative pressure
High economic pressure	Type A: Pure poverty Income policy eliminates marriage.	
Low economic pressure	Type B: Neither Marriage unlikely.	

Mechanisms: the 2×2 Typology

	Low normative pressure	High normative pressure
High economic pressure	Type A: Pure poverty Income policy eliminates marriage.	Type C: Reinforced Income policy partially reduces.
Low economic pressure	Type B: Neither Marriage unlikely.	Type D: Pure norm Income policy has zero effect. Law may or may not bind.

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Modeling Choices

- **No formal bride price or dowry**: economic channel operates through consumption and labor.
- **Adult age heterogeneity**: match is characterized by age gap $\Delta = a_{\text{adult}} - a$ and economic transfer $w(\Delta)$.
 - **Large-gap** ($\Delta \geq \bar{\Delta}$): adult's resources drive marriage; power asymmetry; probably would marry formally before 2019.
 - **Small-gap** ($\Delta < \bar{\Delta}$): less unequal relationship; pregnancy or sexual relationship are primary triggers.
- **Pregnancy** $p \in \{0, 1\}$ **is a state variable**: it raises the immediate value of marriage, imposes a waiting cost σ each period unmarried, and, is interacted with the legal minimum age before 2019.
- **Informal marriage**: agents face a margin choice, formal or informal union, modulated by law.
- **Agency is constrained** (Margarido 2024, Veiga & Zanello 2020): decision is joint family–girl, not pure parental.
- **Norms vary intergenerationally**: usable as identification variation; norms may also vary in their tolerance of large age gaps.

Discrete-Choice Marriage Timing Model

Household i with daughter age $a \in \{10, 11, \dots, 17\}$. In each period decides: *marry formally* ($M = F$), *marry informally* ($M = I$), or *wait* ($M = W$). State is a pair (a, p) , $a \in \{10, \dots, 17\}$, $p \in \{0, 1\}$. Match offer is a pair $(\Delta, w(\Delta))$ drawn from market distribution $F_{\Delta}(a)$.

The utility to get married formally is:

$$U_i^F(a, p, \Delta) = w(\Delta) + \kappa_i N_i(\Delta) + \mu_i R(a) - \phi^F(a, p, t) + \varepsilon_{ia}^F$$

- $w(\Delta)$: economic transfer from groom; $\frac{dw}{d\Delta} > 0$
- $N_i(\Delta)$: norm weight on formal marriage, $\frac{dN}{d\Delta} \leq 0$
- κ_i : weight on norms
- $R(a)$: reputational risk increasing in age; μ_i : sensitivity of reputation;
- $\phi^F(a, p, t)$: registry refusal on the *formal* margin; eliminated by the pregnancy exception before 2019

$$\phi^F(a, p, t) = \bar{\phi}^F \cdot 1[a < \bar{a}_t] \cdot [1 - (1 - 1_{t \geq 2019}) \cdot p]$$

Discrete-Choice Marriage Timing Model

The utility to get married informally is:

$$U_i^I(a, p, \Delta) = w(\Delta) + (\kappa_i - \lambda) N_i(\Delta) + \mu_i R(a) - \phi^I(a) + \varepsilon_{ia}^I$$

- $w(\Delta)$: economic transfer from groom; $\frac{dw}{d\Delta} > 0$
- $N_i(\Delta)$: norm weight on formal marriage, $\frac{dN}{d\Delta} \leq 0$
- κ_i : weight on norms
- $\lambda > 0$: norms slightly penalize informal relative to formal union
- $R(a)$: reputational risk increasing in age; μ_i : sensitivity of reputation;
- $\phi^I(a)$: criminal liability on the *informal* margin; present in all periods and not modified by Lei 13.811.

$$\phi^I(a) = \bar{\phi}^I \cdot 1[a < \bar{a}_t]$$

Discrete-Choice Marriage Timing Model

The utility to not get married and wait is:

$$U_i^W(a, p) = Y_i + s(a) - \sigma p + \delta \mathbb{E}_{p'}[V(a + 1, p')] + \varepsilon_{ia}^W$$

- $s(a)$: schooling return
- Y_i : household income
- σ : cost of pregnancy
- δ : discount factor

① **Composition of pre-16 marriages before 2019:**

- Large- Δ : economic motive ($w(\Delta)$ large) clears the threshold even with $\phi' > 0$; concentrated in poorer households with high norm sensitivity.
- Small- Δ : pregnancy exception ($\phi^F = 0$ when $p = 1$) was the enabling condition; these marriages should vanish on the formal margin post-2019.

② **Pregnancy $\times$ law interaction:** closing the exception raises ϕ from 0 to $\bar{\phi}$ for pregnant girls under 16. Sharpest compliance effect: small- Δ , $p = 1$, $a < 16$ after 2019.

③ **Differential substitution by adult type:**

- Large- Δ : $w(\Delta)$ sustains marriage motive; substitution from formal to *informal* union expected.
- Small- Δ : lower $w(\Delta)$; waiting, and marrying at 18, becomes relatively attractive; less substitution to informal.

④ **Age bunching:** discontinuity at 16 concentrated in small- Δ matches post-2019; large- Δ matches should show bunching in *informal* registries instead.

⑤ **Testable signature of the coercion channel:** post-2019, the formal-to-total union ratio should fall more for large- Δ matches than for small- Δ matches.

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- 1 **Descriptive layer:** Estimate marriage-market model parameters using pre-2019 equilibrium; establish baseline mechanism shares, including age-gap distribution and pregnancy rates at union.
- 2 **Mechanism quantification:** Heterogeneous-effects regression of PBF on marriage, stratified by norm proxy and groom age-gap category.
- 3 **Lei 13.811/2019:** Causal effects on formal and informal margins, disaggregated by Δ and pregnancy status.

Layer 1: Structural Estimation of Mechanism Weights

Estimate discrete-time model of marriage on PNADC 2012–2019, conditioning on state (a, p) and adult type Δ :

$$P(M_{it} = j \mid a_{it}, p_{it}, \Delta_{it}, Y_{it}, N_i, X_{it}) = \frac{\exp(V_{it}^j)}{\sum_k \exp(V_{it}^k)}$$

with V^j from the extended utility specification. Yields:

- $\hat{w}'(\Delta)$: slope of economic transfer in age gap; identifies the **economic channel**.
- $\hat{\sigma}$: pregnancy stigma cost in the waiting payoff; identifies the **pregnancy channel**.
- $\hat{\phi}^I$: implied criminal-liability deterrence on the informal margin.
- $\hat{\kappa}(\Delta)$, $\hat{\mu}$: norm weight and reputational sensitivity.

Counterfactual simulations:

- Raise Y to poverty line $\rightarrow$ norm share.
- Remove large-gap matches $(\Delta \geq \bar{\Delta}) \rightarrow$ economic share.
- Remove pregnancy stigma $(\sigma = 0) \rightarrow$ pregnancy share.

Layer 2: Mechanism Quantification via Heterogeneous Effects

Extensive margin — who marries:

$$P(\text{Union}_i) = \alpha_m + \beta_1 \text{PBF}_i + \beta_2 \text{MomEarly}_i + \beta_3 (\text{PBF}_i \times \text{MomEarly}_i) + \beta_6 p_i + \gamma Y_i + X_i' \delta + \varepsilon_i$$

Intensive margin — age gap conditional on marriage:

$$\Delta_i = \alpha + \beta_4 \text{PBF}_i + \beta_5 \text{MomEarly}_i + \gamma Y_i + X_i' \delta + \varepsilon_i$$

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Intensive margin — age gap conditional on marriage:

$$\Delta_i = \alpha + \beta_4 \text{PBF}_i + \beta_5 \text{MomEarly}_i + \gamma Y_i + X_i' \delta + \varepsilon_i$$

Predictions:

- $\beta_3 > 0$: PBF effect is smaller in high-norm families
- $\beta_4 < 0$: PBF reduces age gap among those who marry
- $\beta_6 > 0$: pregnancy raises marriage propensity

Layer 2: A Norm Index

Build a composite index $\hat{N}_i$:

- Mother's age at first union < 16 (PNADC / PNS)
- Any older sister married before 18
- Share of women 25+ in census tract married before 18 (Census 2010)
- Religion (Census / PNS)
- Rural North or Northeast residence (IBGE region)
- **Median age gap at first union in census tract** (Registro Civil / Census)

Layer 3: Causal Identification of Lei 13.811/2019

Three complementary designs, stratified by adult type Δ and pregnancy p :

- 1 **Event study on formal marriages** (Registro Civil, SIDRA/IBGE):

$$y_{at} = \alpha_a + \lambda_t + \sum_{k \neq -1} \theta_k 1[t - 2019 = k] \cdot 1[a < 16] + \varepsilon_{at}$$

Run separately for small- Δ and large- Δ cohorts; the **sharp drop at small- Δ** is the pregnancy-exception closure signature.

- 2 **Age-based DiD**: under-16 (treated) vs. 16–17 (control) $\times$ pre/post 2019, across data sources (formal and informal).
- 3 **Bunching test** : discontinuity in the age-at-first-union density at 16, pre vs. post 2019. Bunching should concentrate in **small- Δ , pregnant** cases, where ϕ^F flipped from 0 to $\bar{\phi}^F$.

Layer 3: Predicted Pattern by Adult Type

The model generates a *joint* prediction across margins and match types:

Match type	Formal	Informal	Mechanism
Small- Δ ($< \bar{\Delta}$)	$\Downarrow\Downarrow$	$\downarrow$	Exception closed; little substitution because ϕ' was already present on informal margin.
Large- Δ ($\geq \bar{\Delta}$)	$\downarrow$	$\uparrow$	$w(\Delta)$ sustains motive; formal blocked so formal-to-informal substitution .

Layer 3: Heterogeneous Law Effects

Interact the treatment with norm index $\hat{N}_i$ and age-gap category:

$$\begin{aligned} y_{irt} = & \alpha_r + \lambda_t \\ & + \theta_1 \text{Post}_t \cdot \text{Under16}_i \\ & + \theta_2 \text{Post}_t \cdot \text{Under16}_i \cdot \hat{N}_i \\ & + \theta_3 \text{Post}_t \cdot \text{Under16}_i \cdot 1[\Delta_i \geq \bar{\Delta}] \\ & + X_{irt}\beta + \varepsilon_{irt} \end{aligned}$$

Run separately with formal and informal union as outcome.

Predictions:

- $\theta_1 < 0$: formal decrease.
- $\theta_2 > 0$: law works less in high-norm regions
- $\theta_3 > 0$ on formal outcome: large- Δ formal marriages decline less — these matches substitute to informal rather than dissolve.
- $\theta_3 > 0$ on informal outcome: large- Δ informal marriages decline less or increase — the substitution effect; identifies the economic channel surviving the law.

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Primary Data Sources

Source	Coverage	Use
PNADC	2012–present, quarterly	Informal + formal unions, age at first union, spouse age , income, PBF status
PNS	2013, 2019	Prevalence, health, norms; pre/post scenarios
CENSO	Universal	Spatial heterogeneity, tract-level norm index, age-gap distribution by tract
REGISTRO CIVIL	Annual	Formal marriages by age of both spouses ; primary DiD outcome; age-gap distribution
SINASC	Annual	Births to under-16 mothers, union status, father's age at birth (?) ; pregnancy channel

Key Variables

- **Outcome:** indicator of union (formal, informal, any) by age-year.
- **Treatment:** age $\times$ calendar year, cutoff at Lei 13.811/2019 (March 2019).
- **Income:** household per-capita income; PBF receipt indicator; value received.
- **Age gap** $\Delta = a_{\text{groom}} - a$ (**central mechanism variable**): observed in Registro Civil (both ages), PNADC (spouse age), SINASC (father's age at birth (?)).
- **Pregnancy** p (**state variable and legal interaction term**): pregnancy at time of union; reconstructed from SINASC birth dates relative to reported union date; interacts with legal-penalty structure pre/post 2019.
- **Norm index** $\hat{N}_i$: mother's age at first union, sister unions, area early-union share, median age gap in area, religion, race/ethnicity, region.
- **Controls:** household composition, parents' education, race, region, urban/rural, municipal GDP and HDI.

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Next Steps: Immediate (0–1 months)

① Stylized-fact figures:

- Age-specific marriage hazard by year (2010–2024), pre and post 2019.
- **Age-gap distribution at first union for under-16 brides** (Registro Civil and SINASC), tests the large- Δ composition hypothesis directly.
- **Pregnancy rate at union by age** (SINASC), pre and post 2019, establishes pregnancy-trigger baseline.

② **Norm index**: construct and validate $\hat{N}_i$ on PNADC.

Next Steps: Medium (1–5 months)

- 1 **Structural estimation**: discrete-choice hazard model on pre-2019 PNADC, conditioning on (a, p, Δ) . Recover $\hat{w}'(\Delta)$, $\hat{\sigma}$, $\hat{\phi}^I$.
- 2 **Heterogeneous PBF effects**
- 3 **Law evaluation**

Next Steps: Longer Horizon (5–8 months)

- ① **Secondary outcomes:** SINAN marital-rape trends pre and post law; SINASC births-in-union trends. Test whether welfare gains concentrate in small- Δ cases rather than large- Δ cases.
- ② **Policy simulation:** use the structural model to simulate:
 - Complementary norm campaigns on top of the law, targeted at high- $\hat{N}$ regions.
 - **Adult-side deterrence:** simulate increasing ϕ^I , criminal-liability enforcement on the informal margin, for large- Δ matches.

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